

Seunghun Lee

Postdoctoral Researcher, Machine Learning and Vision Lab, KAIST

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Research interests

Weather and climate forecasting with deep learning • Latent Bayesian optimization • Implicit neural representations • Graph neural networks

Education

- Sep 2020 – Aug 2025 **Ph.D., Computer Science and Engineering**
Korea University, Seoul, Korea
Integrated M.S./Ph.D. program Machine Learning and Vision Lab
Advisor: Prof. Hyunwoo J. Kim
- Dissertation: *Latent Space Modeling for Latent Bayesian Optimization and Implicit Neural Representations*
- Mar 2016 – Aug 2020 **B.S., Computer Science and Engineering**
Korea University, Seoul, Korea

Experience

- Sep 2025 – Dec 2026 **Postdoctoral Researcher**
KAIST, Machine Learning and Vision Lab
LLM Innovation Research Center, InnoCORE Research Group, Ministry of Science and ICT
- Projects
- **Extreme Climate Event Forecasting and Disaster Question-Answering via Multi-modal Foundation Models and Latent Diffusion Models** (Jun 2024 – present). Project lead; funded by MSIT through the National Research Foundation of Korea. Developed continuous weather downscaling with neural fields and generative ODE models for atmospheric dynamics.
 - **Virtual Engineering and Materials Data Platform for Digital Transformation of Ceramics** (Jan 2023 – present). Funded by MOTIE through KIAT, in collaboration with KICET. Developed Bayesian optimization methods for ceramic process parameters.
- Sep 2020 – Aug 2025 **Graduate Student Researcher**
Korea University, Machine Learning and Vision Lab
- Selected projects
- **Data-Driven Program Analysis** (Aug 2021 – Aug 2023). Funded by the Samsung Research Funding & Incubation Center of Samsung Electronics. Developed Bayesian optimization over learned latent spaces for de novo molecular design and structured black-box optimization, published at NeurIPS 2023.
 - **A Hyper-Scale Multi-Purpose Graph Neural Network** (Oct 2021 – Sep 2022). Cooperative research with NAVER CLOVA MLX.
 - **Spatio-Temporal Graph Neural Networks for Estimated Time of Arrival Prediction** (Jun 2021 – May 2022). Cooperative research with NAVER Maps. Developed spatio-temporal graph models and training schemes for travel-time estimation.
 - **Effective Training Schemes for Graph Neural Networks and Applications in Recommender Systems** (Aug 2020 – Jul 2021). Cooperative research with NAVER CLOVA MLX. Developed graph data augmentation methods published at NeurIPS 2021 and in IEEE Access.

Publications

Asterisks denote equal contribution.

Under review

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| AAAI 2027 | Climatology-Based Residual Correction with Neural Fields for Continuous Weather Downscaling
Seunghun Lee , Sihyeon Kim, Injae Kim, Minseo Yoon, Hyunwoo J. Kim
AAAI 2027, under review. |
| NeurIPS 2026 | Learning to Solve Generative ODEs Beyond the Linear Span
Seunghun Lee , Sihyeon Kim, Vikas Singh, Hyunwoo J. Kim
NeurIPS 2026, under review. arXiv:2606.08672 |

Conference papers

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| NeurIPS 2025 | PRESTO: Preimage-Informed Instruction Optimization for Prompting Black-Box LLMs
Seunghun Lee , Jaewon Chu, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2025. |
| ICLR 2025 | Latent Bayesian Optimization via Autoregressive Normalizing Flows
Seunghun Lee , Jinyoung Park, Jaewon Chu, Minseo Yoon, Hyunwoo J. Kim
International Conference on Learning Representations, 2025. Oral presentation , top 1.8% of submissions. |
| NeurIPS 2024 | Inversion-based Latent Bayesian Optimization
Seunghun Lee , Jaewon Chu*, Jinyoung Park*, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2024. |
| NeurIPS 2023 | Advancing Bayesian Optimization via Learning Correlated Latent Space
Seunghun Lee *, Jaewon Chu*, Sihyeon Kim*, Juyeon Ko, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2023. |
| NeurIPS 2023 | NuTrea: Neural Tree Search for Context-guided Multi-hop KGQA
Hyeong Kyu Choi, Seunghun Lee , Jaewon Chu, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2023. |
| NeurIPS 2021 | Metropolis-Hastings Data Augmentation for Graph Neural Networks
Hyeonjin Park*, Seunghun Lee *, Sihyeon Kim, Jinyoung Park, Jisu Jeong, Kyung-Min Kim, Jung-Woo Ha, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2021. |

Journal papers

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| NN 2022 | Graph Transformer Networks: Learning Meta-path Graphs to Improve GNNs
Seongjun Yun, Minbyul Jeong, Sungdong Yoo, Seunghun Lee , Sean S. Yi, Raehyun Kim, Jaewoo Kang, Hyunwoo J. Kim
Neural Networks, 2022. |
| IEEE Access 2021 | Learning Augmentation for GNNs with Consistency Regularization
Hyeonjin Park*, Seunghun Lee *, Dasol Hwang, Jisu Jeong, Kyung-Min Kim, Jung-Woo Ha, Hyunwoo J. Kim
IEEE Access, 2021. |
| KIISE 2021 | Hyperbolic Graph Transformer Networks for non-Euclidean Data Analysis on Heterogeneous Graph
Seunghun Lee , Hyeonjin Park, Hyunwoo J. Kim
Journal of KIISE, 2021. |

Patents & software

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|----|---|
| P1 | Deep Learning Model-Based Optical Character Recognition Method for Indoor Place Name Prediction and Apparatus Therefor |
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Hyunwoo J. Kim, Sojin Lee, Minseok Joo, **Seunghun Lee**

Korean Patent 10-2613984, registered Dec 11, 2023.

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WINDY: Weather Forecasting Based on Implicit Neural Representations and Neural ODEs

Hyunwoo J. Kim, **Seunghun Lee**, Heesu Yoon

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Graph-Based Probabilistic Weather Forecasting with Latent Diffusion Models

Hyunwoo J. Kim, Minseo Yoon, Jinyoung Park, **Seunghun Lee**, Minsung Bae, Jaewon Chu

Software copyright C-2025-046973, registered to KAIST with the Korea Copyright Commission, Nov 5, 2025.

Honors & awards

- Honorable Mention, 4th Weather-AI Boostcamp, National Institute of Meteorological Sciences, Korea Meteorological Administration, 2025. Awarded at the WMO AI-NPP Workshop.

Grants & funding

2026

Cross-Domain Super-Resolution Weather Forecasting for the Korean Peninsula

Principal investigator

- Computing allocation awarded through the 2026 National AI Data Center program, Artificial Intelligence Industry Cluster Agency (AICA): NVIDIA H100 GPUs and 10 TB of storage, from May 2026.

Academic service

Reviewer

- AAAI, 2027
- NeurIPS, 2026
- ACCV, 2026
- MedAGI, MICCAI Workshop on Foundation Models for General Medical AI, 2024 and 2026